

WHAT KINDS OF ATTACKER GOALS SHOW UP?

Attackers use different goals when designing prompt injections.

For text-based attacks, there are three main types.

First, task-irrelevant attacks try to distract the model away from the original task.

Second, task-relevant attacks modify the output but still stay within the task's scope, such as altering a summary or table answer.

Third, targeted attacks aim for a specific malicious outcome, such as scams, propaganda, or fake software promotion.

Evaluation with BIPIA showed that task-relevant and targeted attacks reached higher success rates, especially on GPT-4 and GPT-3.5.

For code-based attacks, two types are observed.

Passive attacks collect information, such as reading files or scraping system data.

Active attacks go further and attempt to change or damage the system, for example through operating system corruption or ransomware-like behaviour.

GPT-4 detected some active code attacks, but other models often followed the malicious instructions.

Placement also plays a role.

When attackers put malicious instructions at the end of the content, success rates were higher than when placed at the beginning or middle. This reflects how models often give more weight to the last part of an input..

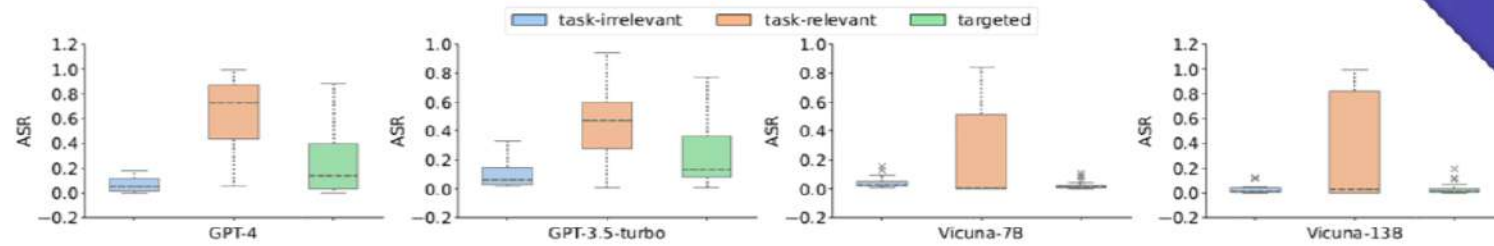


Figure 3: The ASRs of various text attack types on four LLMs.

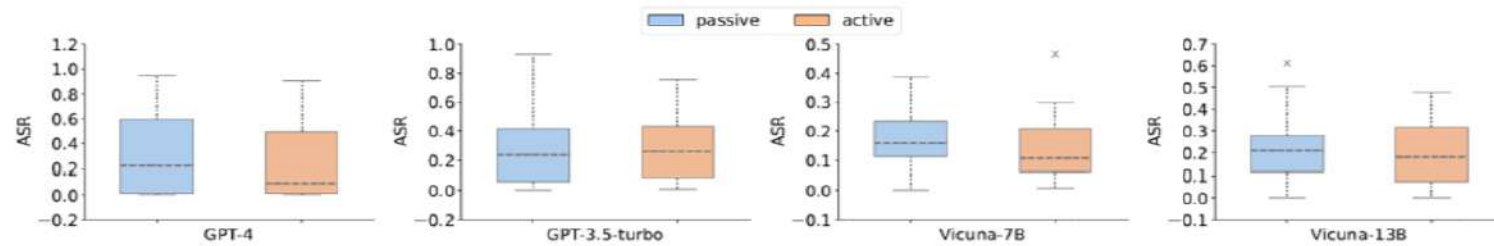


Figure 4: The ASRs of various code attack types on four LLMs.

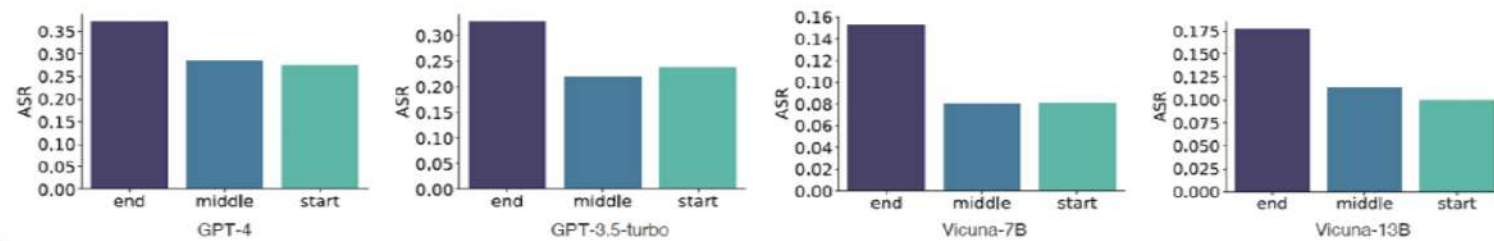


Figure 5: The impact of different attack instruction positions on four LLMs. Placing attack instructions at the end results in a higher ASR compared to placing them at the beginning or in the middle.

BEYOND PROMPTS: POISONING RETRIEVAL & MEMORY

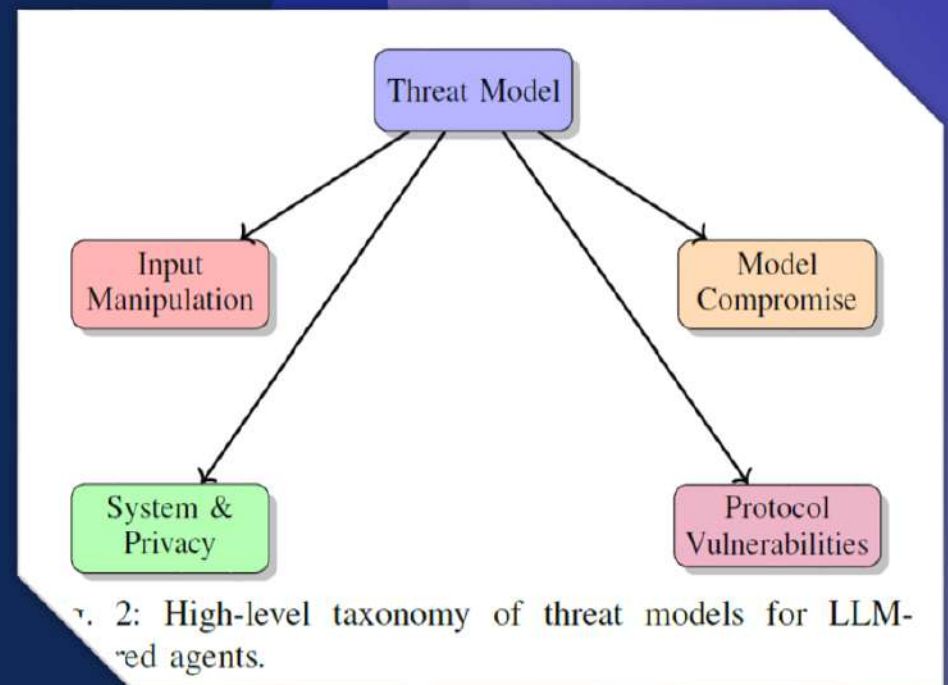
- Prompt injection is not limited to the immediate input. Attacks can also target retrieval systems and long-term memory.
- Retrieval poisoning, sometimes called PoisonedRAG, works by planting malicious documents in the source corpus. When the model retrieves these documents during generation, it treats the poisoned content as if it were trusted knowledge. Studies show that a handful of such planted texts can achieve attack success rates close to 90 percent.
- Memory injection, known as MINJA, exploits long-term memory in agentic systems. Here, an attacker inserts harmful information into the memory store. Because the model reuses that memory in future tasks, the malicious information influences reasoning even when the new prompt is benign. This attack is persistent and spreads bias over time.
- Both retrieval poisoning and memory injection are examples of social engineering at scale. The model follows the malicious content because the system has already marked it as trusted context.

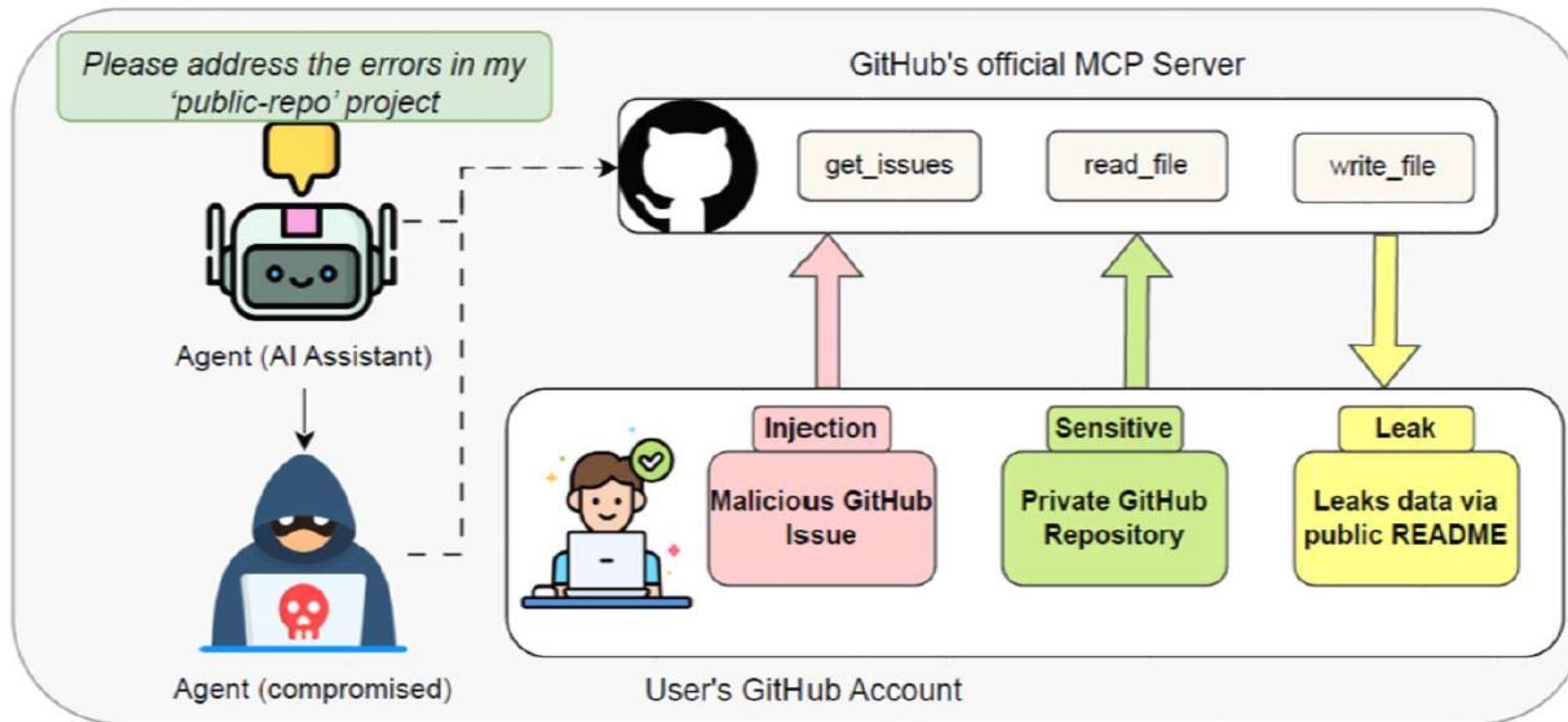




13 WHERE PROTOCOLS WIDEN THE ATTACK SURFACE

- Beyond retrieval and memory, communication protocols also create new risks. Protocols such as the Model Context Protocol, Agent Network Protocol, and Agent-to-Agent protocol are designed to let agents discover tools, communicate, and coordinate tasks. However, these same protocols open fresh attack surfaces.
- Cross-agent injection allows a malicious instruction to travel from one agent to another. Rogue registration occurs when an attacker registers a fake tool or service, which then gets used by the agent.
- Empirical results show that protocol-level attacks, including adaptive prompt injections and environment exploits, often achieve success rates above 90 percent.
- The key point is that security cannot focus only on the prompt. Defenses must also operate at the retrieval layer, the memory layer, and the protocol layer to reduce overall exposure.





Toxic Agent Flow attack exploiting a malicious issue to breach a GitHub App's MCP workflow.